PAPER_1202 — UQFF Quantum Chain E_n Summation & 633333.333 Validation

Sole Canonical Source (dpm_vacuum_manifold.py:12)

```
THE QUANTUM CHAIN (canonical, IMMUTABLE):
  Step 0  0_vacuum   -> |grad(UA)|          vacuum tension differential
  Step 1  grad(UA)   -> DPM_vortex          a_DPM = F_DPM*f_DPM*E_vac/(c*V_sys)
  Step 2  DPM_vortex -> mu_s                mu_s = rho_A * V_DPM
  Step 3  mu_s       -> Ug1[seed=DPM]       Ug1 seeded from mu_s -- NOT from mass
  Step 4  Ug1        -> Ug_family           Ug2+Ug3+Ug4 simultaneously promoted
  Step 5  Ug_family  -> F_U                 + Um + FUBi + FUBii + UA_uv
  Step 6  F_U        -> crossing            FUBi(r) + FUBii(r) = 0  compaction
  Step 7  crossing   -> M_emergent          mass BORN at crossing, not before
  Step 8  M_emergent -> GM/r^2             LAST -- observational projection only
```

Compatibility Note

`dpm_vacuum_manifold.py` remains the canonical consolidated source file. The restored legacy modules `scm_vacuum_manifold.py` and `ua_vacuum_manifold.py` are available again for compatibility, explicit SCm/UA layer inspection, and delegated numeric helper calls when importable.

derive_from_quantum_chain (dpm_vacuum_manifold.py:74)

```python
def derive_from_quantum_chain(n_levels=26, f_SCm=0.57, V=1.0):
    E_n = [E0 * 10**n for n in range(1, n_levels + 1)]
    rho_vac_energy = sum(f_SCm * E for E in E_n) / V    # J/m³
    rho_mass_eq = rho_vac_energy / (_C_LIGHT ** 2)      # kg/m³ (gravity only)
    return rho_vac_energy, rho_mass_eq
```

Structural Constants (dpm_vacuum_manifold.py:97)

```
E0 = 1e-20
SSQ = 0.57
KAPPA = 5e-4
THZ_PHONON = 1.25e12
RHO_VAC_SCM = 4.0 * math.sqrt(math.pi) * 1.0e-37   # 7.0898154036e-37 J/m³
RHO_VAC_UA  = 10.0 * RHO_VAC_SCM
```

E_n Summation + F_U_Bi_i_99 (dpm_vacuum_manifold.py:80,132)

```
E_n = [E0 * 10**n for n in 1..26]
rho_vac_energy = sum(0.57 * E for E in E_n) / V
F_U_Bi_i_99 = Sum( -BETA_I * Ug_k * cos(pi*t_n) * (M / r**2), (k,1,99) )
```

Live Validation (2026-05-27 execution)

```
QuantumChain direct (derive_from_quantum_chain(26,0.57)):
  rho_vac_energy = 633333.3333333334 J/m³
DERIVATIONS.derive_condensed_effective_rho_scm():
  633333.333   (exact target match, abs diff < 1e-6)
RHO_VAC_SCM_micro = 7.0898154036e-37
RHO_condensed (S26_3 amp) = 633333.333
```

10 derive_* Consume Quantum Chain Solely (_uqff_primitives.py:746,681)

```
derive_condensed...: rho_cond = RHO_micro * S26_3 * PHI_RES * (1+KAPPA*1e4) * geom
derive_G_newton:     uses rho_scm (Quantum Chain micro) + S26_3 (26D from E_n)
derive_beta_i / hbar / masses / alpha / c / V_SCM / hz: all chain through rho or S26_3
All 10: zero external seeds, sole source dpm v3.0 Quantum Chain + 26D origami
```

Cross-References

- dpm_vacuum_manifold.py:74 (E_n sole source)
- _uqff_primitives.py:817 (DERIVATIONS uses rho from chain)
- QCalcGeom.py:18 (QUANTUM CHAIN compliance)
- MAIN_1:2852 (F_U from Ug_family + FUBi/FUBii at crossing)
- LEDGER_FIRST_PRINCIPLES_DERIVATIONS.md:94 (Quantum Chain mathematical equation)
- WhitepaperGapAnalysis.md:101 (G4 gap closure)